FINAL PROJECT TITLE: LIBRARY MANAGEMENT SYSTEM



A Project Documentation Submitted to the

College of Information Technology

Cagayan de Oro College

PHINMA Education

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INTRODUCTION

Lumbia National High School currently faces challenges in managing its library operations due to its reliance on a manual system. Recording borrowed and returned books, tracking inventories, and maintaining borrower information often leads to delays, errors, and inefficiency. These challenges not only affect the productivity of the library staff but also inconvenience students who rely on the library for their academic needs. In today's digital age, manual processes can no longer keep up with the growing demand for quick and accurate information management.

To address these issues, our group decided to develop a Library Management System (LMS) that aims to modernize and automate the school's library processes. The system is designed to replace the traditional paper-based workflow with a computerized solution that enhances accuracy, efficiency, and accessibility. It provides features such as user registration, borrowing and returning transactions, and inventory monitoring. By adopting this system, the library staff can easily manage book records, students can quickly check book availability, but they need to approach the librarian to borrow and return materials.

The Library Management System will greatly benefit the students and staff of Lumbia National High School by providing quicker and more reliable access to library resources. With the system in place, students can locate available books more easily, while librarians can efficiently manage transactions and maintain up-to-date records. This improvement promotes a smoother library experience, encourages reading habits among students, and supports their academic growth by ensuring that learning materials are properly managed and readily available.

This system also helps the school move forward with technology, even though modern resources are sometimes hard to come by. It's a big step toward making things more efficient and up-to-date. Because the LMS takes care of the routine work, librarians will have more time to help students and make the library an even better learning environment.

In conclusion, implementing the Library Management System will transform the operations of Lumbia National High School's library from manual to automated. It will significantly reduce human error, save time, and ensure better organization of library resources. Overall, this project aims to create a more accessible, accurate, and productive environment for both students and staff, strengthening the school's commitment to effective learning and resource management.

Statement of the Problem

Lumbia National High School currently faces difficulties in managing its library operations due to its manual record-keeping system. The existing process of tracking borrowed and returned books, monitoring inventory, and maintaining borrower information is time-consuming and prone to human error. This manual method often results in misplaced records, inaccurate data, and inefficiencies that hinder the smooth operation of the library.

The lack of technological resources in the school also further contributes to the problem, as it restricts the adoption of automated solutions that could improve library management. Librarians spend a significant amount of time handling paperwork and manually searching for records, resulting in slower transactions and a less productive workflow. Consequently, students experience delays in borrowing or returning books, and the library staff faces difficulties in tracking book availability and generating reports efficiently. These issues hinder the library's efficiency and make it difficult for both students and staff to access accurate and updated information.

- Manual record-keeping often results in lost or duplicated files, making information retrieval slow and unreliable.

- Borrowing and returning transactions are time-consuming and prone to human error, reducing the accuracy of records.

- Tracking book availability and borrower information is difficult, leading to inefficiency and confusion in operations.

- The absence of an automated system limits the library's ability to monitor, update, and organize data effectively.

- Limited access to modern technology prevents the implementation of a faster, more reliable, and user-friendly library system.

Project Objectives

The main objective of this project is to design and implement a computerized Library Management System (LMS) for Lumbia National High School to replace its traditional, paper-based processes. The proposed system aims to improve the efficiency, accuracy, and organization of library operations by automating key tasks such as book tracking, student registration, and transaction management. Through this system, the library will be able to manage its resources more effectively, reduce human errors, and provide faster service to students and staff.

Specific Objectives

- To digitize library records for easier storage, retrieval, and management of books and borrower information.
- To automate the borrowing and returning processes to minimize human error and reduce processing time.
- To provide real-time inventory tracking that allows librarians to monitor book availability and status efficiently.
- To create a secure and user-friendly interface accessible only to authorized users, such as the library admin.
- To generate accurate reports on borrowed and returned books, helping staff make informed decisions regarding inventory and usage.
- To promote digital transformation within Lumbia National High School by introducing an efficient, technology-driven system despite limited access to modern tools.

Scope of the System

User Management

The system will have a detailed database of users containing their name, ID number, and login credentials. It helps in tracking the students' and teachers' library activities, such as borrowing and returning books.

Book Management

This enables the librarian to add, update, and archive book records from the system. It includes information such as book title, author, category, and availability status. The system automatically updates the status of each book when borrowed or returned, ensuring that users can see real-time availability. This helps maintain an organized and updated library collection.

Borrowing and Returning

Users can borrow and return books efficiently. The system records the date of borrowing and automatically sets the due date. It also notifies the librarian when a book is overdue. If a user fails to return a book on time or loses it, the system can calculate the corresponding fine based on the number of delayed days. This process ensures accountability and proper management of borrowed materials.

Limitations of the System

Only registered students, teachers, and librarians of Lumbia National High School are permitted to use the system. The librarian serves as the system administrator, with privileges to confirm requests, manage book records, and categorize materials. Outsiders or non-members of the school are not allowed to create accounts or access the library's resources. Furthermore, the system relies on a stable internet connection and local server performance, meaning that connectivity issues or server downtime may temporarily affect access to the system.

Requirements Definition

This section defines the key functional and non-functional requirements of the Library Management System developed for Lumbia National High School. These requirements ensure that the system fulfills the operational needs of the school's library and supports its core workflows effectively.

Functional Requirements:

Authentication

- The system shall allow only one library staff member to log in using a predefined username and password.

  The system shall deny access if the login credentials are incorrect.

Book Management

When the admin adds a new book The system will check whether the book already exists based on its unique identifiers, such as the Book ID than system will check whether the book already exists based on its unique identifiers, such as the Book ID to add new books with the following fields:

  - Title
  - Author
  - Category
  - Subject
  - Quantity

- The system shall display all books in a table, whether it's new or old.
- The system shall merge duplicate book entries and update the quantity.

Student Management

- The system shall allow the admin to register students by name.
- The system shall store and display a list of registered students.

Borrowing Books

The system shall allow the admin to:

- Select a registered student and a book.
- Enter the quantity to borrow.
- Specify the due date.

The system also:

- Validate if the book is in stock.
- Deduct the borrowed quantity from the inventory.
- Display the borrowing record in a table.

Returning Books

The system shall allow the admin to process book returns.

The system shall:

- Update the book quantity
- Record the return date
- Calculate a fine of ₱5 per day if the return is late
- Display returned records in a separate table

USERS

This Library Management System is designed for single-user access, with only one role responsible for operating the system.

Library Administrator (Librarian)

Primary (and only) user of the system

Has full access to all features and functionalities

Logs in with a predefined username and password

Responsible for:

- Adding and managing books
- Registering student borrowers
- Issuing books to students
- Receiving returned books
- Managing fines for late returns
- Monitoring book inventory and student activity

Excluded Users

- Students: Do not have direct access to the system. Their information is managed by the admin.
- Teachers/Other Staff: Also not given access to the library, just admin only.

FUNCTIONAL REQUIREMENTS

FUNCTIONAL REQUIREMENT	DESCRIPTION	USER
Admin Login	Allows the admin to log in using the exclusive admin username and password	Admin
Add Book	Enables the admin to add new books with title, author, category, subject, and quantity	Admin
Register Student	Allows the admin to register students by Name and school ID	Admin
Borrow Book	Allows the admin to issue books to students and record the borrow date and due date	Admin
Return Book	Allows the admin to process book returns and calculate fines for late submissions	Admin
View Inventory	Displays all available books and their current quantities	Admin
View Returned Books	Displays the history of returned books and any fines applied	Admin
View Borrowed Books	Shows a list of currently borrowed books with student names and due dates	Admin
Archive Students	Show Archive student	Admin
Archive Books	Show Archive Books	Admin

ENTITIES AND ATTRIBUTES

ENTITY	DESCRIPTION	POSSIBLE ATTRIBUTES
books1	Represents each book in the library inventory.	id(Pk), title, author, category, quantity
students2	Represents the students who borrow books.	student_id (Pk), first_Name, last_Name, GradeLevel, contact, email_address, address
borrow_books3	Records of students and books after Borrowing.	borrow_id(Pk), student_id, first_name, last_name, book_title, category quantity, borrow_date, due_date, fine
ret_books1	Records of books returned.	id(Pk), first_name, last_name, book_title, quantity, fine, return_date
users1	The user who operates the system.	id(Pk), first_ name, last_name username, password
archive_students1	Show the archive student	id(Pk), student_id, first_name, last_name, grade_level, archived_date, contact, email_addess, address
archive_books	Show the archive student	id(Pk), title, author, category, quantity

REPORTS

REPORT NAME	DESCRIPTION
Book Inventory Report	Shows all the available books along with side with the title, author, category, subject, and quantity
Registered Students	Lists of all students who are registered in the system
Borrowed Books Report	Displays records of the books currently borrowed by students
Returned Books Report	Shows the history of all the returned books, including the return date and fines
Overdue Report	(Optional) it depends if the book is returned after the due date then it could show borrowed books that are past due (if you implement it)

A. Database Design

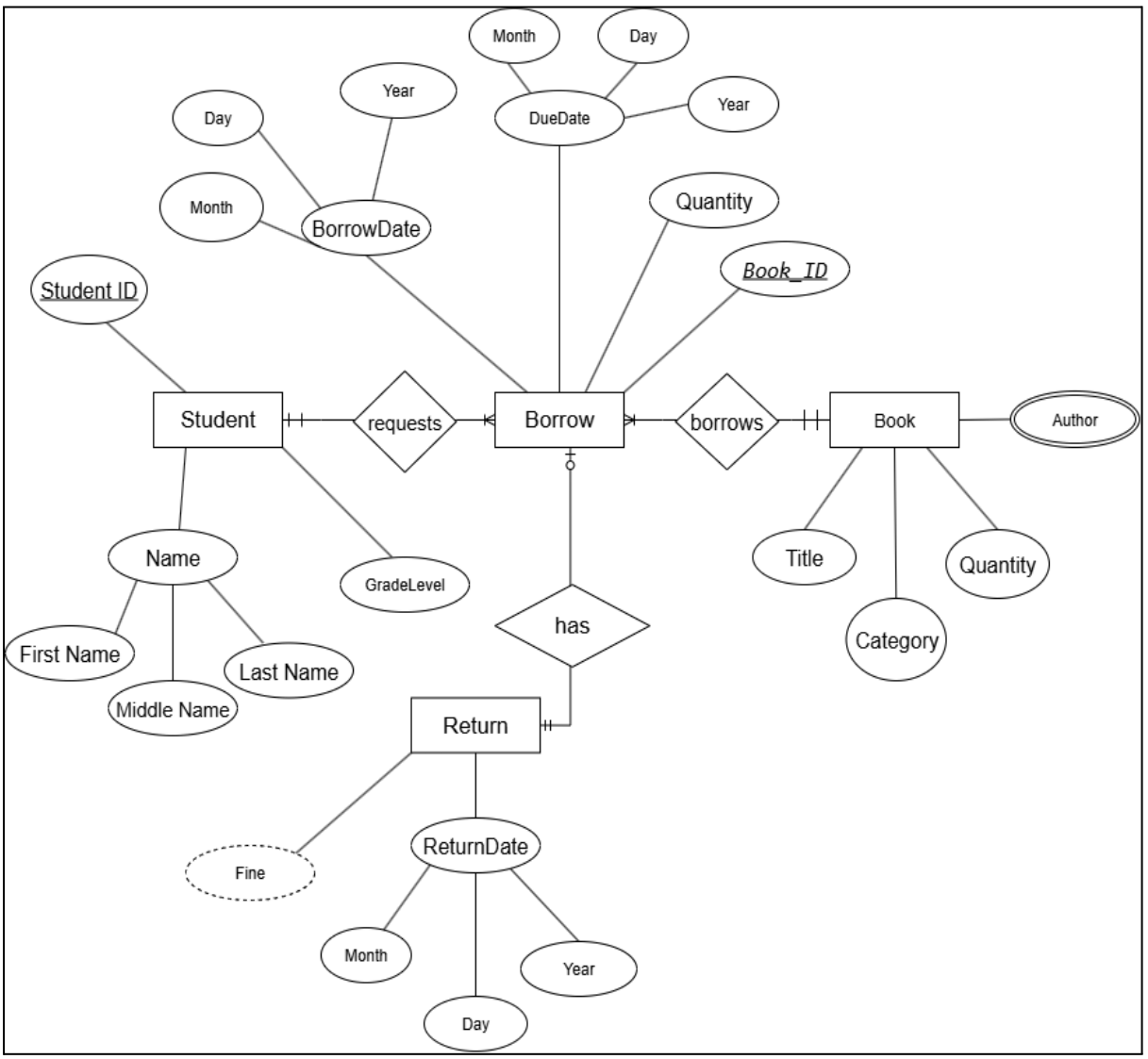


Figure 1: Logical Erd

The Logical Entity Relationship Diagram (ERD) shows how data moves within the Library Management System and how the main entities—Student, Book, Borrow, and Return—are connected. It illustrates that students can borrow and return books, but only the admin can operate the system. The admin manages student records, book details, and borrowing transactions to keep all data organized and accurate.

Data Dictionary

Data Name	Description	Type	Length	Index
id	Unique identifier for Students	INT	11	PK
email	User Email address	VARCHAR	255	
first_name	User Firstname	VARCHAR	100	
last_name	User Lastname	VARCHAR	100	
username	User Username	VARCHAR	100	UNI
password	User Password	VARCHAR	255	

Table 1. User

Data Name	Description	Type	Length	Index
student_id	Unique identifier for Student	CHAR	12	PK
first_name	Students Firstname	VARCHAR	50	
last_name	Students Lastname	VARCHAR	50	
grade_level	Students Grade level	VARCHAR	10	
contact	Students contact	VARCHAR	15	
email_address	Students email address	VARCHAR	100	
address	Students address	VARCHAR	255	

Table 2. Student

Data Name	Description	Type	Length	Index
id	Unique identifier for Books	INT	11	PK
title	Title of the Book	VARCHAR	255	
author	Author of the Book	VARCHAR	255	
category	Book category	VARCHAR	255	
quantity	Number of copies available	INT	11	

Table 3. Books

Data Name	Description	Type	Length	Index
id	Unique identifier for Books	INT	11	PK
title	Title of the Book	VARCHAR	255	
author	Author of the Book	VARCHAR	255	
category	Book category	VARCHAR	100	
quantity	Number of copies available	INT	11	

Table 4. Archive books

Data Name	Description	Type	Length	Index
borrow_id	Unique identifier for Borrowed Book	INT	11	PK
student_id	Unique identifier for Student	VARCHAR	12	MUL
first_name	Student Firstname	VARCHAR	100	
last_name	Student Lastname	VARCHAR	100	
book_Title	Title of the book	VARCHAR	255	
category	Books Category	VARCHAR	100	
quantity	Number of books you get.	INT	11	
borrow_date	The date you borrowed the book.	DATE		
due_date	Due date of borrowed book.	DATE		
fine	Overdue Fee	DECIMAL	10,2	

Table 5. Borrowed books

Data Name	Description	Type	Length	Index
id	Unique identifier for Returned	INT	11	PK
student_id	Unique identifier for Student	INT	12	
first_name	Student Firstname	VARCHAR	100	
last_name	Student Lastname	VARCHAR	100	
book_title	Title of the book	VARCHAR	255	
quantity	Number of books you borrowed.	INT	11	
fine	Overdue Fee	DOUBLE		
return_date	Return date for books	date		

Table 6. Returned books

Data Name	Description	Type	Length	Index
id	Unique identifier for Archive	INT	11	PK
student_id	Unique identifier for Students	INT	12	
first_name	Students Firstname	VARCHAR	100	
last_name	Students Lastname	VARCHAR	100	
grade_level	Students grade_level	VARCHAR	50	
archive_date	Stores archive date	TIMESTAMP	15	
contact	Students number	VARCHAR	100	
email_address	Students email_address	VARCHAR	255	
address				

Table 7. Archive Student

Normalized Tables (3rd BNF)

Table Name	Description	PK	FK	Fields
users1	Stores all user information.	id		id, email, first_name, Last_name, Username, password.
students2	Stores all students information	student_id	-	student_id, firstName, lastName, gradeLevel, contact, email_address, address
archive_students1	Stores archive students	id		id, student_id, first_name, last_name, Grade_level, Contact, archive_date, email_address address
books1	Stores all books information	book_id	-	book_id, title, author, category, quantity
archive_books	Stores archive books	id		book_id, title, author, category, quantity
borrowed_books3	Stores borrowing transactions between students and books	borrow_id		borrow_id, student_id, book_id, first_name, last_name, book_title, category, quantity, borrow_date, due_date, fine

ret_books1	Stores returned books	id		id, student_id, first_name , last_name, book_title, quantity, fine, returned_date

Case Diagram

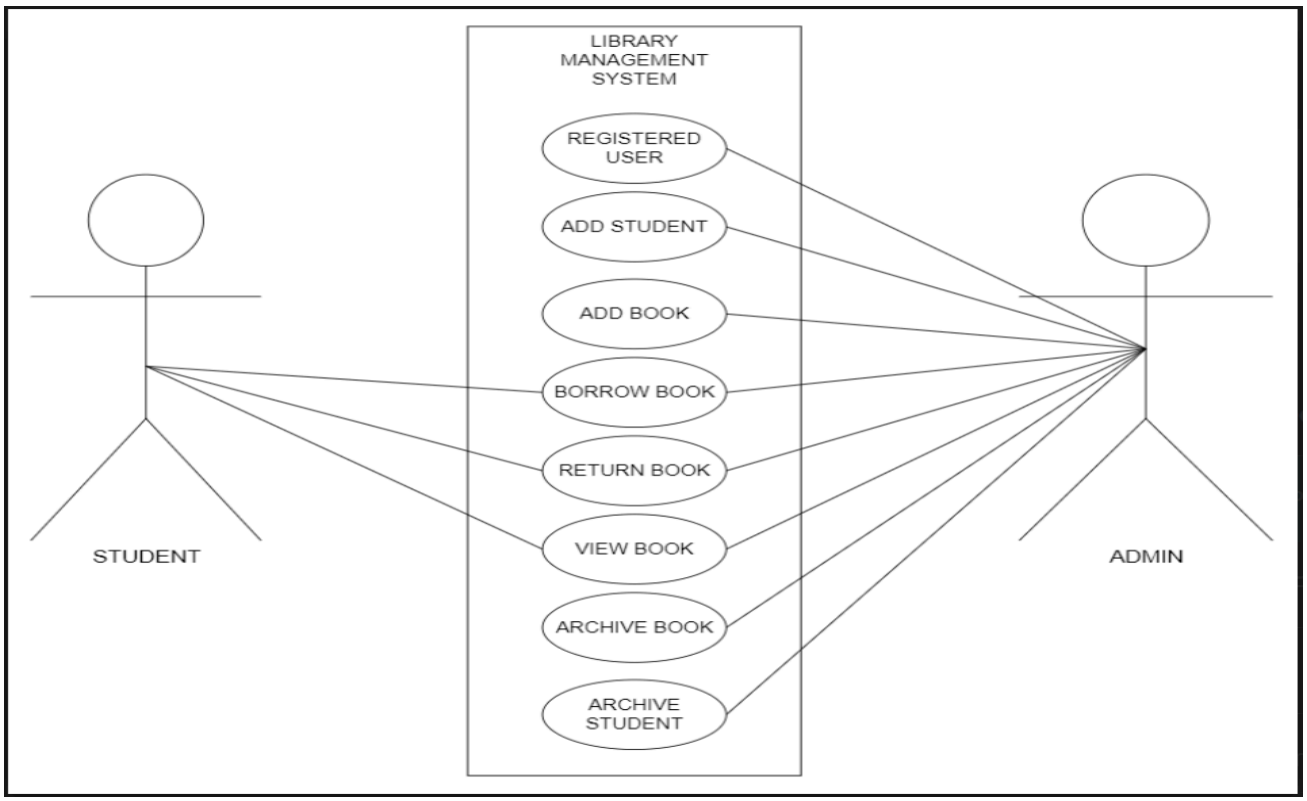


Figure 2: Use Case Diagram

The Use Case Diagram shows how data flows within the Library Management System and who can use it. In this system, only the admin has direct access to operate the system. The student cannot log in or use the system on their own. Instead, the admin is responsible for registering students, adding books, and managing all library operations. Once a student is registered by the admin, the admin can record and process the borrowing and returning of books on behalf of the student. This means all transactions—such as adding, borrowing, returning, and archiving books—are controlled and handled by the admin. The student interacts with the system indirectly, by requesting the librarian (admin) to borrow or return books. This setup ensures data accuracy and security, as only the admin can modify or manage the system's records.

Context Diagram

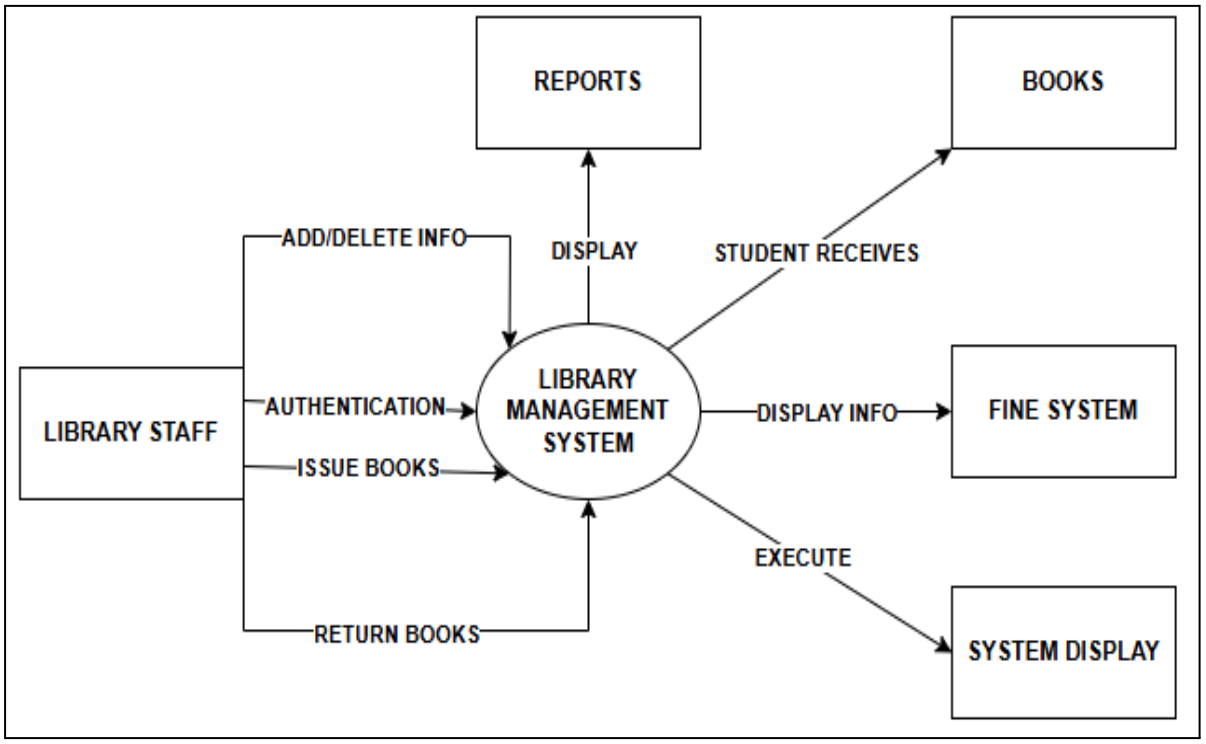


Figure 3: Context Diagram

The Context Diagram shows the overall flow of information in the Library Management System and how it interacts with external entities. It illustrates that only the library staff (admin) can access and manage the system, performing tasks such as adding or deleting information, issuing and returning books, and handling authentication. The system communicates with other components like the Books, Fine System, System Display, and Reports to display and process data efficiently. This setup ensures smooth operation and organized management of library activities.

Physical Database Design & Implementation

1. **Physical Database Design (** Insert Image below**)**

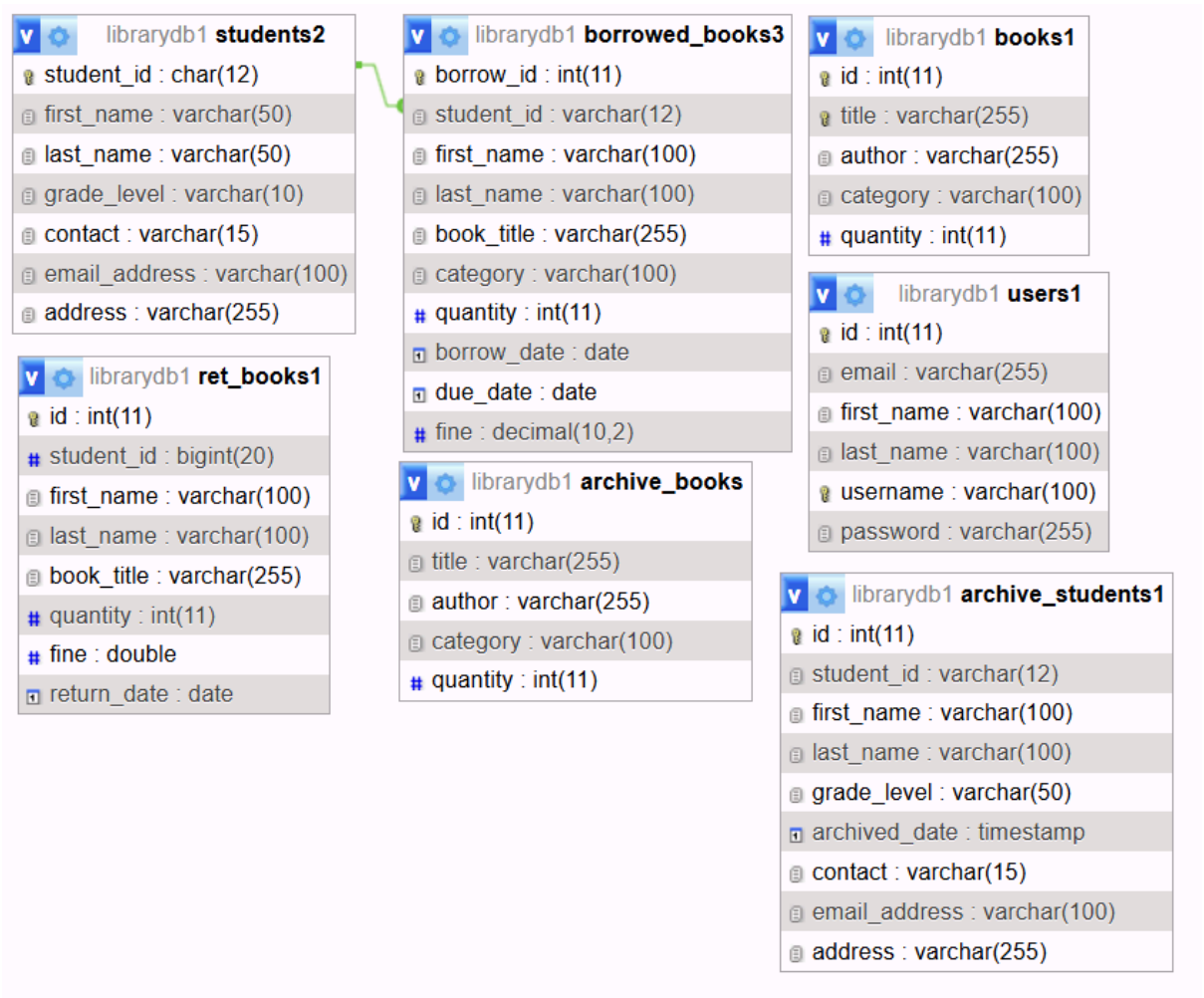


Figure 4: Physical Database design

The Physical Database Design illustrates the actual structure of the Library Management System database, showing how tables are organized and related. It includes tables such as students2, books1, borrowed_books3, archive_books, users1, ret_books1, and archive_students1, each storing specific information about students, books, transactions, and users. This design ensures that data is properly stored, retrieved, and managed, allowing the admin to efficiently track student information, borrowed and returned books, and archived records within the system

Table Structures (Actual database**)**

```
DESCRIBE users1;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	*NULL*	auto_increment
email	varchar(255)	NO		*NULL*	
first_name	varchar(100)	NO		*NULL*	
last_name	varchar(100)	NO		*NULL*	
username	varchar(100)	NO	UNI	*NULL*	
password	varchar(255)	NO		*NULL*	

Figure 5. User Table

```
DESCRIBE students2;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
student_id	char(12)	NO	PRI	*NULL*	
first_name	varchar(50)	NO		*NULL*	
last_name	varchar(50)	NO		*NULL*	
grade_level	varchar(10)	NO		*NULL*	

Figure 5. Students Table

```
DESCRIBE archive_students1;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	*NULL*	auto_increment
student_id	varchar(12)	NO		*NULL*	
first_name	varchar(100)	NO		*NULL*	
last_name	varchar(100)	NO		*NULL*	
grade_level	varchar(50)	NO		*NULL*	
archived_date	timestamp	NO		current_timestamp()	

Figure 6. Archive Students Table

```
DESCRIBE books1;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	*NULL*	auto_increment
title	varchar(255)	NO	UNI	*NULL*	
author	varchar(255)	NO		*NULL*	
category	varchar(100)	NO		*NULL*	
quantity	int(11)	NO		0	

Figure 7. books Table

```
DESCRIBE archive_books;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	*NULL*	auto_increment
title	varchar(255)	NO		*NULL*	
author	varchar(255)	NO		*NULL*	
category	varchar(100)	YES		*NULL*	
quantity	int(11)	YES		1	

Figure 8.Archive Books Table

```
DESCRIBE borrowed_books3;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
borrow_id	int(11)	NO	PRI	*NULL*	auto_increment
student_id	varchar(12)	NO	MUL	*NULL*	
first_name	varchar(100)	NO		*NULL*	
last_name	varchar(100)	NO		*NULL*	
book_title	varchar(255)	NO		*NULL*	
category	varchar(100)	YES		*NULL*	
quantity	int(11)	NO		*NULL*	
borrow_date	date	NO		*NULL*	
due_date	date	NO		*NULL*	
fine	decimal(10,2)	YES		0.00	

Figure 9.Borrowed Books Table

```
DESCRIBE ret_books1;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	*NULL*	auto_increment
student_id	bigint(20)	NO		*NULL*	
first_name	varchar(100)	YES		*NULL*	
last_name	varchar(100)	YES		*NULL*	
book_title	varchar(255)	NO		*NULL*	
quantity	int(11)	NO		1	
fine	double	YES		0	

Figure 10: Returned Books Table

SQL Queries

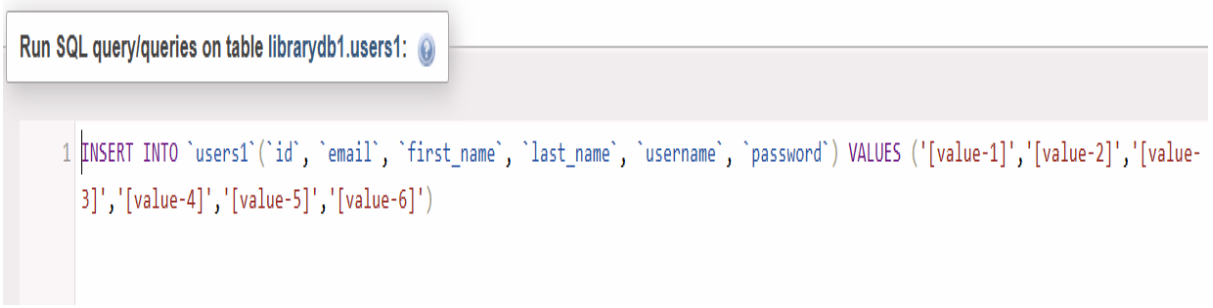


Figure 11: 1. User Insertion Query Result

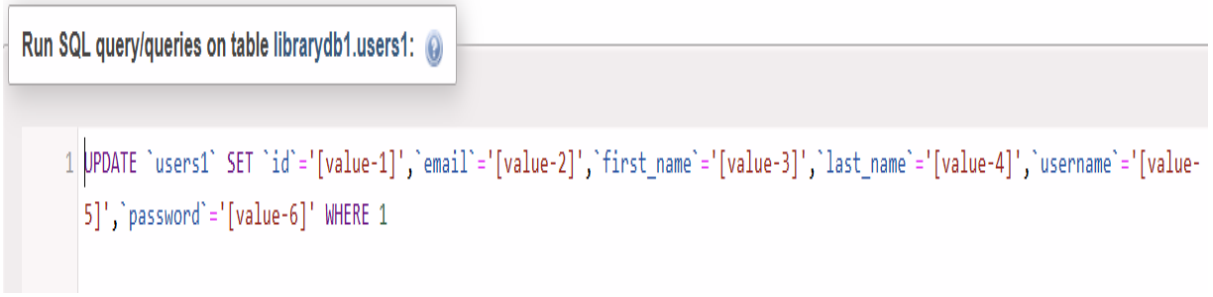


Figure 11. 2 .User Update Query Result

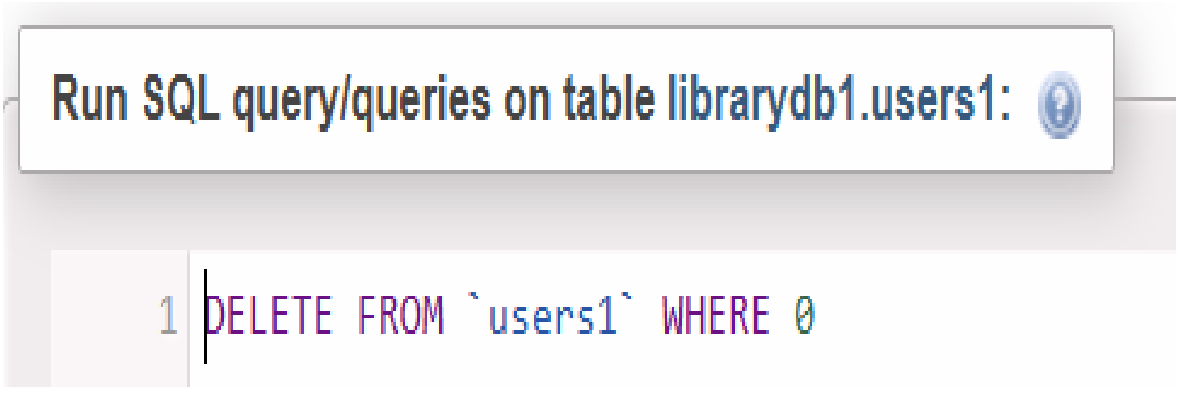


Figure 11: 3. User Deletion Query Result

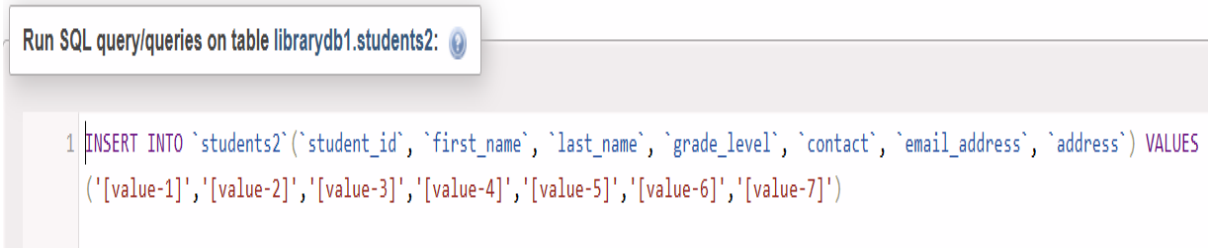


Figure 11: 4. Student Insertion Query Result



Figure 11: 5. Student Update Query Result

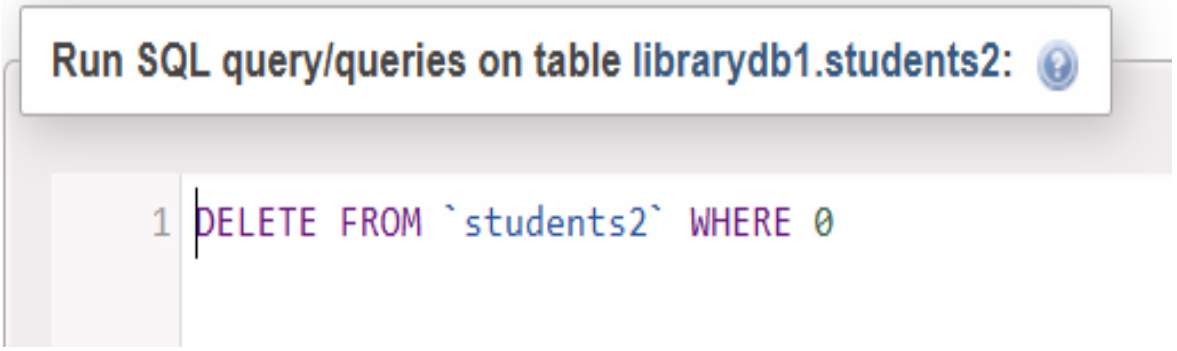


Figure 11: 6. Student Deletion Query Result

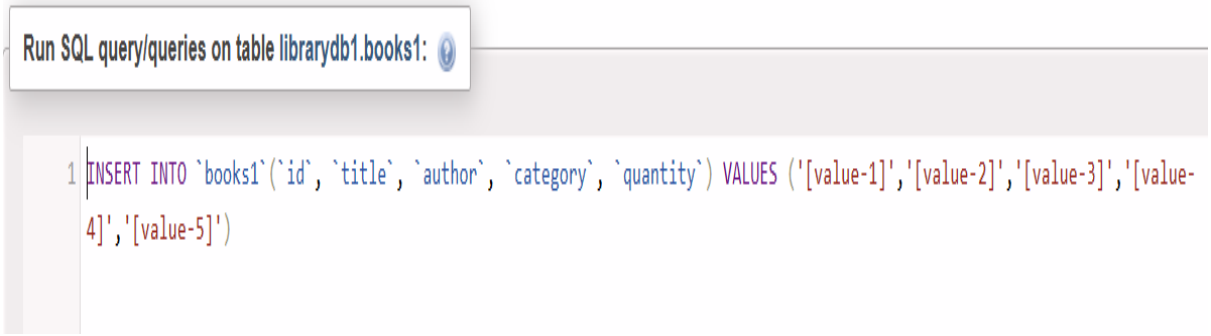


Figure 11: 7. Books Insertion Query Result



Figure 10: 8. Books Update Query Result

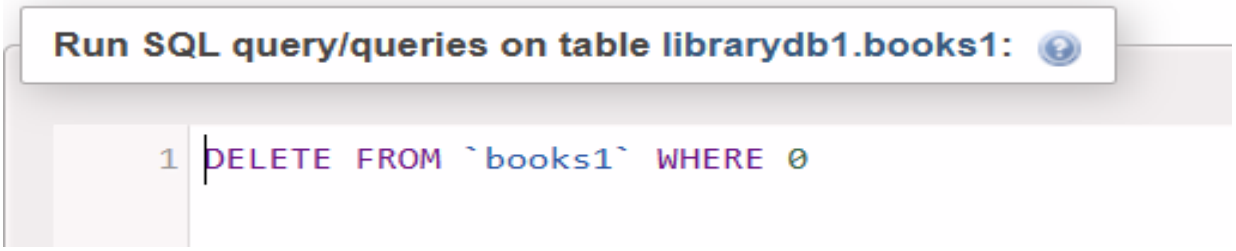


Figure 11: 9. Books Deletion Query Result

Run SQL query/queries on table librarydb1.borrowed_books3:

```
INSERT INTO `borrowed_books3`(`borrow_id`, `student_id`, `first_name`, `last_name`, `book_title`, `category`, `quantity`, `borrow_date`, `due_date`, `fine`) VALUES ('[value-1]','[value-2]','[value-3]','[value-4]','[value-5]','[value-6]','[value-7]','[value-8]','[value-9]','[value-10]')
```

Figure 11: 10. Borrowed Books Insertion Query Result

Run SQL query/queries on table librarydb1.borrowed_books3:

```
UPDATE `borrowed_books3` SET `borrow_id`='[value-1]',`student_id`='[value-2]',`first_name`='[value-3]',`last_name`='[value-4]',`book_title`='[value-5]',`category`='[value-6]',`quantity`='[value-7]',`borrow_date`='[value-8]',`due_date`='[value-9]',`fine`='[value-10]' WHERE 1
```

Figure 10: 11. Borrowed Books Update Query Result

Run SQL query/queries on table librarydb1.borrowed_books3:

```
DELETE FROM `borrowed_books3` WHERE 0
```

Figure 11: 12. Borrowed Books Deletion Query Result

Run SQL query/queries on table librarydb1.ret_books1:

```
1 INSERT INTO `ret_books1`(`id`, `student_id`, `first_name`, `last_name`, `book_title`, `quantity`, `fine`, `return_date`) VALUES
('[value-1]','[value-2]','[value-3]','[value-4]','[value-5]','[value-6]','[value-7]','[value-8]')
```

Figure 11: 13. Return Books Insertion Query Result

Run SQL query/queries on table librarydb1.ret_books1:

```
1 UPDATE `ret_books1` SET `id`='[value-1]',`student_id`='[value-2]',`first_name`='[value-3]',`last_name`='[value-
4]',`book_title`='[value-5]',`quantity`='[value-6]',`fine`='[value-7]',`return_date`='[value-8]' WHERE 1
```

Figure 10: 14. Return Books Update Query Result

Run SQL query/queries on table librarydb1.ret_books1:

```
1 DELETE FROM `ret_books1` WHERE 0
```

Figure 11: 15. Return Books Deletion Query Result

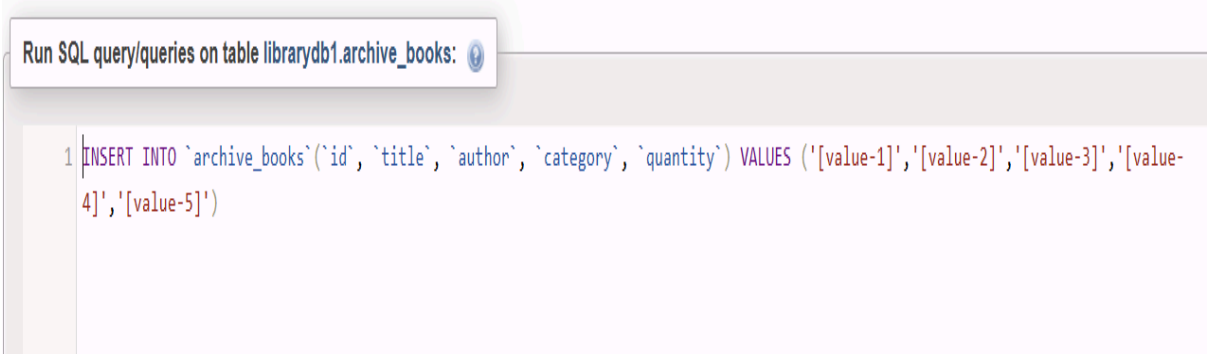


Figure 11: 16. Archive Books Insertion Query Result

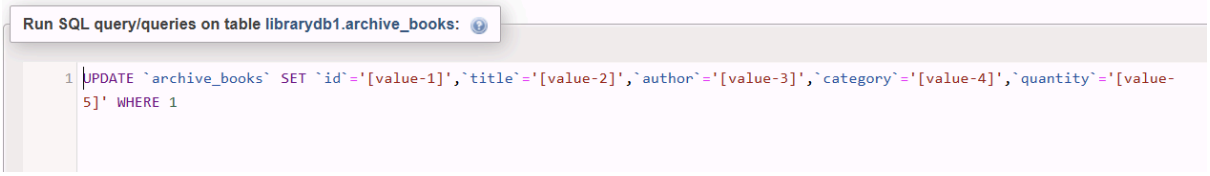


Figure 11: 17. Archive Books Update Query Result



Figure 11: 18. Archive Books Deletion Query Result

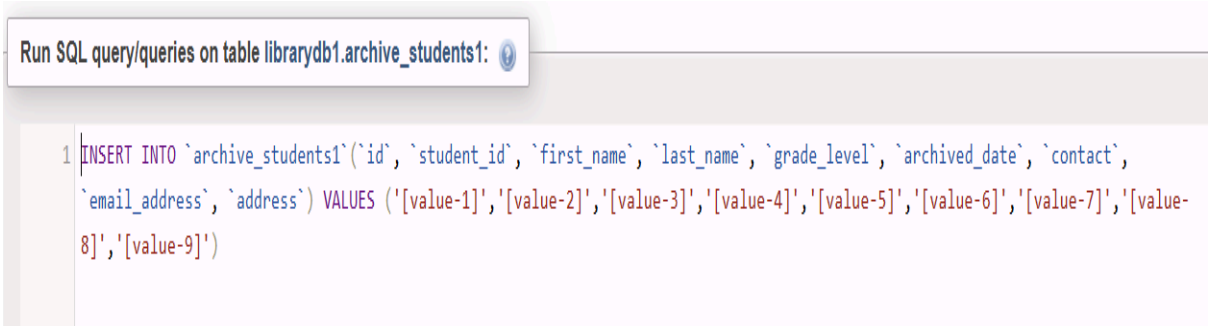


Figure 11: 19. Archive Books Insertion Query Result



Figure 10: 20. Archive Books Update Query Result

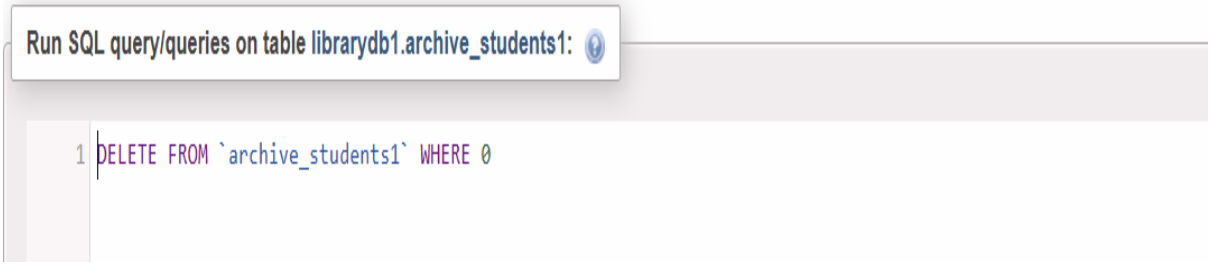


Figure 11: 21. Archive Books Deletion Query Result

Reports Design
SUMMARY, FINDINGS, AND RECOMMENDATION

Summary

The proposed Library Information System for Lumbia National High School is designed to replace the school's manual library operations with a digital system. It aims to organize and centralize records of books, students, and borrowing transactions in one place that can only be accessed by the admin. The system includes features for managing student registration, book information, borrowing, returning, and archiving. Through this setup, the system is expected **to** improve efficiency, accuracy, and record management once implemented.

Findings

Based on the database design and system planning, the proposed Library Information System is capable of addressing the common issues found in the manual process, such as misplaced records, duplication, and time-consuming searches. The normalization of the database up to the Third Normal Form (3NF) ensures data accuracy and reduces redundancy. The system allows the admin to handle all records and transactions effectively, making book tracking and student management more organized. However, since the system has not yet been implemented, further testing and user evaluation are still needed to confirm its full functionality and performance.

Recommendation

It is recommended that once the system is implemented, the librarian and staff undergo proper training on how to use and manage the system efficiently. Regular system orientation and practice sessions should be conducted to ensure familiarity with the features and data management processes. The librarian should also be advised to maintain regular data backups and perform periodic system checks to prevent data loss and ensure smooth operation. Additionally, feedback from the librarian can be gathered after initial use to improve the system further and adapt it to the actual workflow of the library.